# BioSpecInfo

## Software Bill of Materials (SBOM)

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Versione **bsi-v209** · commit `b5a356a`

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Documento generato automaticamente dal sorgente Markdown versionato nel repository. La versione e il commit stampati in ogni piè di pagina identificano esattamente lo stato del software descritto.

# Software Bill of Materials (SBOM)

**BioSpecInfo** · document generated by `tools/genera-sbom.js`.

| | |
|---|---|
| Application version | `bsi-v209` |
| Commit | `b5a356a0a5cf8497fc752085553bb0d84465c3a6` |
| Generated (UTC) | `2026-10-15T13:46:59.945Z` |
| Machine-readable format | `../evidence/sbom.cdx.json` — CycloneDX 1.5 |

> The SHA-256 digests refer to the files actually distributed in this commit. A reviewer can recompute them with `sha256sum <file>` and compare, confirming they hold the very artefact described here.

## 1. Third-party components distributed with the application

Every component is **included in the repository** and served from the same origin: the application loads no code from external CDNs at runtime. That is what makes it usable offline and what removes any dependence on third-party availability.

| Component | Version/File | Licence | Size | Role |
|---|---|---|---|---|
| **RDKit MinimalLib (JS glue)**<br><sub>RDKit / Greg Landrum</sub> | `RDKit_minimal.js` | BSD-3-Clause | 125 kB | Cheminformatics: SMILES and SMARTS parsing, descriptors, 2D drawing. |
| **RDKit MinimalLib (WebAssembly)**<br><sub>RDKit / Greg Landrum</sub> | `RDKit_minimal.wasm` | BSD-3-Clause | 6753 kB | WebAssembly binary of the cheminformatics engine. |
| **three.js**<br><sub>mrdoob e collaboratori</sub> | `three.min.js` | MIT | 589 kB | WebGL rendering engine for the 3D scenes. |
| **three.js — UnrealBloomPass**<br><sub>mrdoob e collaboratori</sub> | `three_bloom.js` | MIT | 25 kB | Post-processing effect for the astronomical visualisations. |
| **three.js — GLTFLoader**<br><sub>mrdoob e collaboratori</sub> | `gltf_loader.js` | MIT | 94 kB | Loading of anatomical and molecular models in glTF format. |
| **3Dmol.js**<br><sub>University of Pittsburgh / David Koes</sub> | `3Dmol-min.js` | BSD-3-Clause | 502 kB | 3D molecular visualisation (PDB, SDF, surfaces). |

| Component | Version/File | Licence | Size | Role |
|---|---|---|---|---|
| **SmilesDrawer**<br><sub>Daniel Probst</sub> | `smiles-drawer.min.js` | MIT | 246 kB | 2D structure drawing from SMILES. |
| **PDF.js (build legacy)**<br><sub>Mozilla Foundation</sub> | `vendor/pdfjs/pdf.min.mjs` | Apache-2.0 | 511 kB | Opening PDF documents in the spectrum reader: every page drawn and its text layer. It is the legacy build, because the default one uses very recent language features and breaks on browsers that do not have them yet. |
| **PDF.js — worker**<br><sub>Mozilla Foundation</sub> | `vendor/pdfjs/pdf.worker.min.mjs` | Apache-2.0 | 1290 kB | The PDF decoding work, off the main thread: a complex page does not freeze the interface. |
| **sql.js (JS glue)**<br><sub>sql.js / SQLite</sub> | `lib/sql-wasm.js` | MIT | 45 kB | In-browser SQLite database for the data exercises. |
| **SQLite (WebAssembly)**<br><sub>SQLite Consortium</sub> | `lib/sql-wasm.wasm` | Public Domain | 644 kB | SQLite engine compiled to WebAssembly. |
| **Dataset dimuoni (CERN Open Data)**<br><sub>CERN Open Data Portal</sub> | `lib/dimuon.js` | CC0-1.0 | 48 kB | Experimental data for the particle-physics exercise. |

## Digests

| File | SHA-256 |
|---|---|
| `RDKit_minimal.js` | `58d3c996ade7e0b0137d4f9363ff6c204b545689428fa0898eeaed579ef788d9` |
| `RDKit_minimal.wasm` | `e0967d44fed59e44a2d07bfc08f3a86821c54a14c95bb7e6e392fe867333b8c8` |
| `three.min.js` | `9274bbcec8d96168626c732b5d31c775aa8cfb7eaa0599bec0c175908a2c1ce2` |
| `three_bloom.js` | `42e0ff40b00ea1ba4fe53baeb3726834a3ac3db46e9cfb80856baebf01611090` |
| `gltf_loader.js` | `5c15967ba830918a9caea6338712c994c354bccd4edc4569bde411c3ec06a3e6` |
| `3Dmol-min.js` | `06b6d2fc7d418e8bef62a32cca44373ff21b2b787bd377613dd4039394a4e9ff` |
| `smiles-drawer.min.js` | `955285fac52a7017b3c3658e2cb84bef7f1f76732f3f7cd565d1bb0903232dc2` |
| `vendor/pdfjs/pdf.min.mjs` | `bccc24ea711db8e44503629519904a5292d73b9daaa214bbe7cdcc282b0f4259` |
| `vendor/pdfjs/pdf.worker.min.mjs` | `145d2dd3ab0c86151011dba95acfa2d5336e2accd59388ea43dbee0efddaaec6` |
| `lib/sql-wasm.js` | `77d6435bac506af0e3c59636dce9d22b1b14156348bc327f41a1577f3212360f` |
| `lib/sql-wasm.wasm` | `438c88f666dc054ce4e9395f80fe9db4218b1a3c379960454880f048a7898aed` |
| `lib/dimuon.js` | `61190d7bbec3ce4ea3add2503b9513ac27b227a3878dd229560a6cd29aabc36d` |

## 2. Code written for this project

| File | Size | Role |
|---|---|---|
| `index.html` | 4670 kB | Main application: interface, teaching sections, chemical data. |
| `bsi-ai-hub.js` | 422 kB | «Spectra» AI agent: agentic loop, tools, provider management. |
| `bsi-spettri.js` | 35 kB | IR/NMR spectral prediction engine over the molecular graph. |
| `bsi-cheminfo.js` | 125 kB | Cheminformatics engine: standardisation, fingerprints, clustering, PCA, QSAR models with a null model. |
| `bsi-pretsch.js` | 37 kB | The NMR estimation tables transcribed in full from Pretsch–Bühlmann–Badertscher, 4th ed.: $^{13}$C and $^{1}$H benzenes, ethylenes, alkanes, alkynes, $^{13}$C aliphatics, steric corrections, $^{1}$J(C,H). Numbers only, no code. |
| `bsi-nmr.js` | 57 kB | Per-atom $^{1}$H and $^{13}$C NMR prediction: applies the bsi-pretsch.js tables by walking the ring and measuring distances, with a cyclic reference compound and chemical equivalence by environment code. |
| `bsi-geom3d.js` | 46 kB | 3D coordinates from a SMILES by distance geometry, carrying the NMR predictor's own atom indices: the precondition for the peak↔atom↔3D link. |
| `bsi-nmr2d.js` | 21 kB | Two-dimensional NMR: COSY, edited HSQC and HMBC, built by walking the graph over the two per-atom predictions. |
| `bsi-moldraw.js` | 44 kB | Interactive NMR spectrum: zoom, assignment table, peak↔atom link, SMILES/InChI history, CSV/JCAMP-DX/PNG exports. |
| `bsi-documento.js` | 23 kB | Opening of any document — PDF, images, text, Word, OpenDocument, PowerPoint, Excel — with an explicit list of what could NOT be read. |
| `bsi-quesito.js` | 19 kB | Recognition of spectroscopic data in a problem's text, with each datum's provenance, and step-by-step working through bsi-elucida.js. |
| `bsi-spettrolettore.js` | 104 kB | Spectrum reader: JCAMP-DX with ASDF compression, prominence-based peak finding, IR band assignment, neutral losses. |
| `bsi-mol3d.js` | 42 kB | 3D molecule viewer: formation animation, atom selection, measurement of angles, lengths and dihedrals. |
| `bsi-lingue.js` | 163 kB | Internationalisation engine: fourteen languages, writing direction included. |
| `bsi-molingue.js` | 54 kB | Chemical languages: twenty-five outputs from SMILES, with IUPAC naming over a declared class. |
| `bsi-pannelli-lingua.js` | 66 kB | Panels for languages and chemical languages. |
| `sw.js` | 6 kB | Service Worker: offline operation and network strategy. |
| `rdkit_lab.html` | 260 kB | Cheminformatics laboratory. |
| `astro.html` | 2660 kB | Astrochemistry module. |

| File | Size | Role |
|---|---|---|
| `chimorga.html` | 194 kB | Organic chemistry module. |
| `proxy/spectra-proxy.js` | 10 kB | Optional proxy (Cloudflare Worker) for the AI calls. |

### Digests

| File | SHA-256 |
|---|---|
| `index.html` | `cd6666855146376983097b36bcb0529b57514a76df3e3b02c4989ed7b25445d3` |
| `bsi-ai-hub.js` | `985b9fa3f6a3d330cc6f7e31456e16760d0a0d9482f760b04d83765a013f2476` |
| `bsi-spettri.js` | `be757f15a70f884e3374d401596e2dfd18bcc852cafbb4584920e2b2d24eb3c9` |
| `bsi-cheminfo.js` | `59a3111b3b9436add5376fde1be6f90ded3dc08c095cebdc04da50c7f8268bbc` |
| `bsi-pretsch.js` | `27f727240363579fba3dba50ab3b567fbcdb26a8a9a06c0e43968b195c479db3` |
| `bsi-nmr.js` | `881917dbb141dfcbab9a25b9c518eaa4a8c1c52dc1020cbac90985a1cf859b72` |
| `bsi-geom3d.js` | `a616b91efa8f790388c99f22cc9f2d48630cb760d4f7fa87f92b99b9db106054` |
| `bsi-nmr2d.js` | `a0b1db87f27ffa8c124d617e4903f61e35b89c30c35162527ad023325c03fa78` |
| `bsi-moldraw.js` | `45cacdca20c8091cbfb32e256c179de8c32c71068a5d1008fbef9be005b5df0e` |
| `bsi-documento.js` | `0e9ad6043a396652549b30879764cc401f8b7bd25689b5bb5f37cf26f7d7d8d9` |
| `bsi-quesito.js` | `6f686abcfdc3959d8c865a3e6ee7129224de52992ab1ba3da07372a986769cdf` |
| `bsi-spettrolettore.js` | `14daf41305b0a7271c6ff5f7c9078a4e5f0e0149f32460a9608ed1996f810256` |
| `bsi-mol3d.js` | `93fea4a6811ba4263de450db4e7ae1801d4fde196cf7683d0100846c51751893` |
| `bsi-lingue.js` | `5a28570f930be99f9e07c46ed354dab3392805820209adb1eb38e28f2d20f410` |
| `bsi-molingue.js` | `e45fec1ee20e3e7b9a7bea3d6e20aa4e33e74ec43578bfa7bfb94646b7bc53fb` |
| `bsi-pannelli-lingua.js` | `4af94d70ef104cf21be18ff7fbc7a2248c56e376e2ae48e8cac83878f4554f93` |
| `sw.js` | `32c2ab8a730214e9250ed285492f7c6149e76a0bcae84ba18bdb7c3c1c0f8be5` |
| `rdkit_lab.html` | `066a42092b829df2fa306b691b9d162b6d44edfdb5317ce4cc8a276bc34c80ad` |
| `astro.html` | `3376b2b7f26ed483f2d698c6692b31f21c9f245f87d1fb1884216fb8b1da489d` |
| `chimorga.html` | `047934abc73a79fbaf06ed7036b6e89a913ebdf5354fc52cecdbee7e8f94f51d` |
| `proxy/spectra-proxy.js` | `808c0a12891ae9c6f45cf0c676b033075089e064df9047be844f03badf940962` |

## 3. Development dependencies

Used **only** to run the verification benches. They are not distributed with the application and are not present in production.

| Package | Version | Licence |
|---|---|---|
| `playwright-core` | 1.62.1 | Apache-2.0 |

## 4. Runtime dependencies

**None.** The application has no backend, installs no package at runtime and loads no script from external domains. The only network calls are optional and go to documented public services:

| Service | Purpose | If unreachable |
|---|---|---|
| PubChem PUG-REST | IUPAC name, CAS, GHS pictograms, 3D conformers | the app uses local data and says so |
| NASA / ESA | astronomical images and data | the module shows the tabulated data |
| AI providers (user's choice) | assistant answers | the assistant reports it and offers alternatives |

No personal data is transmitted: API keys, if entered, stay in the device's `localStorage`, or on the user's own proxy (see `proxy/`).

## 5. Licence compatibility

All licences of the distributed components (MIT, BSD-3-Clause, CC0, Public Domain) are **permissive** and mutually compatible. No component is subject to a strong copyleft licence (GPL/AGPL): there is no redistribution constraint on the code written for this project.

The remaining obligations are of **attribution**: the licence notices of the upstream projects are reproduced in `THIRD_PARTY_NOTICES.md`.

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